

Year 7 • Chapter 1 • Integers: answers (teacher copy)

1.01 Integers and opposites (booklet p. 3)

WE DO examples (teacher)

- Ex 1a: -6 ; $+4$
- Ex 1b: 9 , -15 , 0
- Ex 2a: -3 , $+3$; back where it started
- Ex 2b: -8 (8 units from 0)
- Ex 3a: -25
- Ex 3b: 7 and -7

Questions: Easy 1–16, Medium 17–29, Challenging 30–34

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|-----------|-----------|---|
| 1. -55 | 13. 69 | 25. 10 |
| 2. -27 | 14. 61 | 26. 19 |
| 3. -45 | 15. -12 | 27. -30 , 30 |
| 4. 6 | 16. 43 | 28. -39 , 39 |
| 5. 21 | 17. 52 | 29. 83 , -83 |
| 6. 19 | 18. 56 | 30. No, $-0 = 0$ |
| 7. -3 | 19. 45 | 31. -18 |
| 8. -48 | 20. 39 | 32. -2 , -1 , 0 , 1 , 2 |
| 9. -21 | 21. 2 | 33. Yes: every integer has an opposite (0 is its own) |
| 10. 87 | 22. 55 | 34. 11 ; -11 |
| 11. 30 | 23. 20 | |
| 12. -50 | 24. 11 | |

1.02 The number line (booklet p. 4)

WE DO examples (teacher)

- Ex 1a: dots at -5 , -2 , 0 , 3
- Ex 1b: -2 , -1 , 0 , 1
- Ex 2a: dots at -4 , -1 , 2 , 5
- Ex 2b: -5
- Ex 3a: scale in 5s; dots at -30 , -15 , 10
- Ex 3b: 9

Questions: Easy 1–14, Medium 15–30, Challenging 31–35

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|----------|----------|--|
| 1. 14 | 13. 13 | 25. 9 |
| 2. 12 | 14. 13 | 26. -3 |
| 3. 4 | 15. 22 | 27. 3 |
| 4. -3 | 16. 2 | 28. -4 |
| 5. 3 | 17. 36 | 29. 8 |
| 6. 12 | 18. 17 | 30. -5 |
| 7. 3 | 19. 20 | 31. scale -40 to 40 in 10s with dots |
| 8. 2 | 20. 4 | 32. -3 , -2 , -1 , 0 , 1 , 2 |
| 9. 12 | 21. 3 | 33. 17 |
| 10. -1 | 22. 15 | 34. -3 |
| 11. 5 | 23. 7 | 35. 54 |
| 12. 11 | 24. 8 | |

1.03 Comparing and ordering integers (booklet p. 5)

WE DO examples (teacher)

- Ex 1a: False; false
- Ex 1b: -6 , -1 , 0 , 3 , 5
- Ex 2a: $>$
- Ex 2b: 8 , 3 , 0 , -2 , -15
- Ex 3a: -5 , $-(-2)$, 1 , $-(-3)$
- Ex 3b: -11 , -10 , -9

Questions: Easy 1–16, Medium 17–29, Challenging 30–34

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|---------------------------------|---------------------------------|--|
| 1. $>$ | 14. -25 , -12 , -7 , -2 | 27. 22 , 21 , 15 , 3 , -10 |
| 2. $<$ | 15. -13 , -8 , 5 , 12 | 28. 27 , 12 , -24 , -25 , -26 |
| 3. $>$ | 16. -6 , 0 , 2 , 3 | 29. 28 , 20 , 19 , 13 , 2 |
| 4. $<$ | 17. false | 30. e.g. -6 , -7 , -8 |
| 5. $>$ | 18. true | 31. -8 |
| 6. $<$ | 19. false | 32. -1000 , -100 , -10 , -1 , 0 |
| 7. $<$ | 20. true | 33. Yes, for negatives: -20 is smaller than -2 |
| 8. $>$ | 21. false | 34. -4 , $-(-2)$, 3 , $-(-4)$, $-(-9)$ |
| 9. $<$ | 22. true | |
| 10. -10 , -5 , 6 , 8 | 23. true | |
| 11. -15 , -12 , -3 , 10 | 24. false | |
| 12. -16 , 2 , 15 , 19 | 25. true | |
| 13. -21 , -13 , -3 , 14 | 26. true | |

1.04 Adding integers (booklet p. 6)

WE DO examples (teacher)

- Ex 1a: -1
- Ex 1b: -10 ; 4
- Ex 2a: -11
- Ex 2b: 1
- Ex 3a: 5°C
- Ex 3b: -13

Questions: Easy 1–16, Medium 17–27, Challenging 28–32

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|-----------|------------|-------------------------|
| 1. 5 | 12. 2 | 23. 132 |
| 2. 6 | 13. 8 | 24. 125 |
| 3. -2 | 14. 4 | 25. 18 |
| 4. 5 | 15. 5 | 26. 85 |
| 5. 2 | 16. -12 | 27. 6 |
| 6. -4 | 17. 7 | 28. 5°C |
| 7. 6 | 18. -15 | 29. level 5 |
| 8. 5 | 19. 7 | 30. $-\$15$ |
| 9. 0 | 20. -10 | 31. 6 |
| 10. -18 | 21. -18 | 32. -12 |
| 11. -8 | 22. -145 | |

1.05 Subtracting integers (booklet p. 7)

WE DO examples (teacher)

- Ex 1a: -4 ; -12
- Ex 1b: 7 ; 4
- Ex 2a: -12
- Ex 2b: 6
- Ex 3a: 15°C
- Ex 3b: -8

Questions: Easy 1–16, Medium 17–32, Challenging 33–37

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|----------|-----------|--|
| 1. -14 | 14. 2 | 27. 7 |
| 2. -5 | 15. -4 | 28. 22 |
| 3. -11 | 16. 1 | 29. 12 |
| 4. -11 | 17. -25 | 30. 8 |
| 5. -10 | 18. -15 | 31. -13 |
| 6. -18 | 19. 19 | 32. 4 |
| 7. -19 | 20. -1 | 33. 16°C |
| 8. -14 | 21. -13 | 34. 25 m |
| 9. -4 | 22. -48 | 35. -75 m (75 m deep) |
| 10. -6 | 23. 14 | 36. -13 |
| 11. 5 | 24. 56 | 37. -8 |
| 12. 11 | 25. 32 | |
| 13. 16 | 26. 30 | |

1.06 Multiplying integers (booklet p. 8)

WE DO examples (teacher)

- Ex 1a: $-18; -10$
- Ex 1b: $48; -24$
- Ex 2a: -42
- Ex 2b: 40
- Ex 3a: -8
- Ex 3b: Positive: 100 negatives make 50 pairs

Questions: Easy 1–16, Medium 17–30, Challenging 31–35

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|----------|------------|--|
| 1. -18 | 14. $-$ | 27. -6 |
| 2. 9 | 15. $-$ | 28. 9 |
| 3. -40 | 16. $-$ | 29. 5 |
| 4. -42 | 17. -100 | 30. 6 |
| 5. 30 | 18. -20 | 31. -28 m (28 m deep) |
| 6. 80 | 19. -6 | 32. -7 |
| 7. -14 | 20. 4 | 33. Negative: 2 pairs make positive, 1 negative left |
| 8. -18 | 21. -24 | 34. -18°C |
| 9. 48 | 22. -150 | 35. -32 |
| 10. $+$ | 23. -36 | |
| 11. $+$ | 24. -20 | |
| 12. $+$ | 25. -5 | |
| 13. $+$ | 26. -1 | |

1.07 Dividing integers (booklet p. 9)

WE DO examples (teacher)

- Ex 1a: $3; -3$
- Ex 1b: 2
- Ex 2a: -3
- Ex 2b: 8
- Ex 3a: $-\$12$
- Ex 3b: 54

Questions: Easy 1–16, Medium 17–32, Challenging 33–37

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|-----------|----------|------------------------------------|
| 1. 6 | 14. 11 | 27. -2 |
| 2. -6 | 15. -5 | 28. -5 |
| 3. -1 | 16. -6 | 29. -3 |
| 4. 4 | 17. 3 | 30. -8 |
| 5. -7 | 18. -3 | 31. 2 |
| 6. 8 | 19. 4 | 32. -1 |
| 7. -9 | 20. -1 | 33. $-\$21$ |
| 8. -4 | 21. -4 | 34. -3°C each hour |
| 9. -5 | 22. -5 | 35. -35 |
| 10. 8 | 23. 1 | 36. -8 |
| 11. -12 | 24. -1 | 37. -60 |
| 12. -9 | 25. -5 | |
| 13. -2 | 26. -4 | |

1.08 Order of operations with integers

(booklet p. 10)

WE DO examples (teacher)

- Ex 1a: -12
- Ex 1b: -4
- Ex 2a: -18
- Ex 2b: 7
- Ex 3a: $1 \div (-3) = -1/3$
- Ex 3b: 0

Questions: Easy 1–16, Medium 17–24, Challenging 25–30

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|-----------|-----------|----------------------------------|
| 1. 7 | 12. 90 | 23. 0 |
| 2. 35 | 13. -5 | 24. 15 |
| 3. -12 | 14. 0 | 25. -4 |
| 4. 8 | 15. 32 | 26. $-1/3$ |
| 5. 6 | 16. -14 | 27. 4 |
| 6. 6 | 17. -5 | 28. 4 |
| 7. 1 | 18. 26 | 29. $(8 - 12) \div (-4 + 2) = 2$ |
| 8. 3 | 19. 0 | 30. e.g. $-3 \times 4 - 5 = -17$ |
| 9. -29 | 20. -1 | |
| 10. -10 | 21. -9 | |
| 11. 4 | 22. 6 | |

1.09 Integers on a calculator (booklet p. 11)

WE DO examples (teacher)

- Ex 1a: $-9; -6$
- Ex 1b: 5
- Ex 2a: -12
- Ex 2b: 87
- Ex 3a: -52
- Ex 3b: 4

Questions: Easy 1–16, Medium 17–29, Challenging 30–34

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|------------|------------|--|
| 1. -33 | 14. -238 | 27. 85 |
| 2. 397 | 15. 154 | 28. 405 |
| 3. 469 | 16. -400 | 29. $2\ 232$ |
| 4. -167 | 17. 54 | 30. $\approx -1000; -1029$ |
| 5. 667 | 18. -1 | 31. $\approx -20; -20.5$ |
| 6. 19 | 19. 58 | 32. The calculator squares 6 first, then makes it negative. Enter $(-6)^2$ |
| 7. 55 | 20. 12 | |
| 8. -80 | 21. -58 | 33. It should be $8: \times$ is done before $-$ |
| 9. 213 | 22. -27 | |
| 10. 456 | 23. 42 | 34. 5 |
| 11. -170 | 24. 2 | |
| 12. -33 | 25. 24 | |
| 13. -168 | 26. -16 | |

1.10 Integer problems (booklet p. 12)

WE DO examples (teacher)

- Ex 1a: 16°C
- Ex 1b: $-\$45$
- Ex 2a: level -3
- Ex 2b: $\$85$
- Ex 3a: $\$820$
- Ex 3b: -12 m

Questions: Easy 1–11, Medium 12–18, Challenging 19–22

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|-------------------------|-------------|---|
| 1. 1°C | 9. $\$106$ | 17. 104 m |
| 2. 0°C | 10. $\$111$ | 18. 225 m |
| 3. 26°C | 11. $-\$31$ | 19. $26 \times 80 - 6 \times 210 = \820 |
| 4. -6°C | 12. 191 m | 20. $35 - 27 = 8$ |
| 5. 4°C | 13. 138 m | 21. 70 minutes |
| 6. 16°C | 14. 393 m | 22. e.g. It was 4°C and fell 12°C |
| 7. $\$61$ | 15. 341 m | |
| 8. $-\$57$ | 16. 114 m | |